

2081

Bachelor of Science in Computer Science and Information Technology

Course Title: **Advanced Database**

Code No: **CSC461** Semester: **8th Semester**

Full Marks: **60** Pass Marks: **24** Time: **dia database**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Explain different object database concepts in brief. What are type constructors?
2. Why do we need query optimization in databases? Explain heuristic query optimization with example.
3. Explain different distributed database architectures? What is data fragmentation?

Section B

Attempt any Eight questions[8×5=40]

4. Explain different constraints of specialization and generalization.
5. What is hashing? How does hashing improve database efficiency?
6. Explain ODMG object model.
7. What are the uses of query trees in query processing? Explain.
8. What is CAP theorem? Explain.
9. What is Big Data? What are its 3 V's?
10. What are triggers? What are the uses of triggers?
11. Why do we need spatial databases? What are common types of analysis for spatial data?
12. Write short notes on: a) Map Reduce b) Multimedia database